

$$\frac{1}{1+e^{-2x}} = \sum_{n=0}^{\infty} (-1)^n e^{-2nx}$$

$$L(z) = \frac{1}{\Gamma(z)} \sum_{n=0}^{\infty} (-1)^n \int_0^{\infty} x^{z-1} e^{-(1+2n)x} dx$$

$$\int_0^{\infty} x^{z-1} e^{-(1+2n)x} dx = \frac{\Gamma(z)}{(1+2n)^z}$$

$$L(z) = \frac{1}{\Gamma(z)} \sum_{n=0}^{\infty} (-1)^n \frac{\Gamma(z)}{(1+2n)^z} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(1+2n)^z}$$

$$L'(z) = \sum_{n=0}^{\infty} (-1)^n \frac{-\ln(1+2n)}{(1+2n)^z} = - \left[\frac{\ln 1}{1^z} - \frac{\ln 3}{3^z} + \frac{\ln 5}{5^z} - \frac{\ln 7}{7^z} + \frac{\ln 9}{9^z} - \dots \right] =$$

$$= \left(\frac{\ln 3}{3^z} - \frac{\ln 5}{5^z} \right) + \left(\frac{\ln 7}{7^z} - \frac{\ln 9}{9^z} \right) + \dots = \sum_{k=0}^{\infty} \frac{\ln(4k+3)}{(4k+3)^z} - \frac{\ln(4k+5)}{(4k+5)^z} =$$

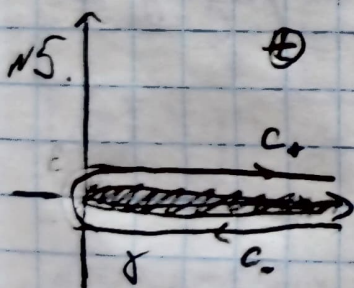
$$= \ln \prod_{n=0}^{\infty} \exp \left(\frac{\ln(4k+3)}{(4k+3)^z} - \frac{\ln(4k+5)}{(4k+5)^z} \right) \quad c_n$$

$$L'(z \rightarrow 0) = \ln \prod_{n=0}^{N-1} \exp(c_n) + \sum_{n=N}^{\infty} c_n \quad ; \quad 1 \ll N \ll e^{1/2}$$

$$\ln \prod_{n=0}^{N-1} \exp(c_n) \rightarrow \ln \prod_{n=0}^{N-1} \frac{4n+3}{4n+5} = \ln \left(\frac{\Gamma(\frac{5}{4})}{\Gamma(\frac{3}{4})} \cdot N^{-\frac{1}{2}} \right)$$

$$\sum_{n=N}^{\infty} c_n \rightarrow \int_N^{\infty} \left[\frac{\ln(4n+3)}{(4n+3)^z} - \frac{\ln(4n+5)}{(4n+5)^z} \right] dn = \ln 2 - \frac{1}{2} \ln \frac{1}{N}$$

$$L'(0) = \ln \left(\frac{\Gamma(\frac{5}{4})}{\Gamma(\frac{3}{4})} \cdot N^{-\frac{1}{2}} \right) + \ln 2 - \frac{1}{2} \ln \frac{1}{N} = \ln \frac{2 \Gamma(\frac{5}{4})}{\Gamma(\frac{3}{4})}$$



$$\int = \int_{C_+} + \int_{C_-} \quad , \quad \int_{C_+} = L(z) \quad ; \quad \int_{C_-} = - \int_{C_+} e^{2\pi i(z-1)}$$

$$\int = (1 - e^{2\pi i z}) \int_{C_+} \Rightarrow L(z) = \frac{1}{1 - e^{2\pi i z}} \frac{1}{\Gamma(z)} \int_{\gamma} \frac{e^{-t} t^{z-1}}{1 + e^{-2t}} dt$$

$$1 + e^{-2t} = 0$$

$$e^{-2t} = -1 = e^{i(\pi + 2\pi n)}$$

$$-2t = i(\pi + 2\pi n)$$

$$t = -\frac{i\pi}{2} - i\pi n = \frac{\pi}{2}(2n+1)e^{\frac{i\pi}{2}}$$

$$\text{res}_{n \geq 0} \frac{e^{-t} t^{z-1}}{1 + e^{-2t}} = e^{-t} t^{z-1} \Big|_{t = \frac{\pi}{2}(2n+1)e^{\frac{i\pi}{2}}} = \frac{(-1)^n e^{-\frac{i\pi}{2}}}{2i} \left(\frac{\pi}{2} \right)^{z-1} (2n+1)^{z-1} e^{\frac{i\pi}{2}}$$

$$\text{res}_{n \leq 0} \frac{e^{-t} t^{z-1}}{1 + e^{-2t}} = e^{-t} t^{z-1} \Big|_{t = \frac{\pi}{2}(2n+1)e^{\frac{i\pi}{2}}} = \frac{e^{-\frac{i\pi}{2}}}{2i} (-1)^n \left(\frac{\pi}{2} \right)^{z-1} (2n+1)^{z-1} e^{\frac{i\pi}{2}}$$

⑥

$$C = \int_0^{\infty} \frac{\ln x dx}{\cosh x}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$L(z) = \frac{1}{\Gamma(z)} \int_0^{\infty} \frac{x^{z-1}}{e^x + e^{-x}} dx = \frac{1}{2\Gamma(z)} \int_0^{\infty} \frac{x^{z-1}}{\cosh x} dx$$

$$2 \frac{d}{dz} [L(z) \Gamma(z)] = \int_0^{\infty} \frac{x^{z-1} \ln x}{\cosh x} dx$$

$$C = 2 \frac{d}{dz} [L(z) \Gamma(z)] \Big|_{z=1} \quad \textcircled{6}$$

$$L(1-z) = \left(\frac{z}{\pi}\right)^z \sin \frac{\pi z}{2} \Gamma(z) L(z)$$

$$C = 2 \frac{d}{dz} \left[L(1-z) \left(\frac{\pi}{2}\right)^z \frac{1}{\sin \frac{\pi z}{2}} \right] \Big|_{z=1} =$$

$$= 2 \left[\left(\frac{\pi}{2}\right)^z \ln \frac{\pi}{2} \cdot \frac{1}{\sin \frac{\pi z}{2}} - \cot\left(\frac{\pi z}{2}\right) \left(\frac{\pi}{2}\right)^{z+1} \frac{1}{\sin \frac{\pi z}{2}} \right] L(1-z) - 2 L'(1-z) \left(\frac{\pi}{2}\right)^z \frac{1}{\sin \frac{\pi z}{2}} \Big|_{z=1}$$

$$= \pi \cdot \ln \frac{\pi}{2} L(0) - L'(0) \cdot \pi = \frac{\pi}{2} \ln \frac{\pi}{2} - \pi \ln \frac{2\Gamma(\frac{5}{4})}{\Gamma(\frac{3}{4})}$$